

Epoxy Cure Oven Specification Rev 4.2

Warmup ramp region: use consecutive segments with both endpoints between 60 C and 120 C.

Maximum warmup ramp: 1.40 C/min.

Effective cure hold: count only time between 150 C and 162 C using linear interpolation at both boundaries.

Batch effective hold is the minimum hold across all embedded thermocouples.

Release window for effective hold: 600 s to 840 s inclusive.

Peak overshoot is $\max(0, \text{hottest peak} - \text{target_peak_c from batch log})$.

Release overshoot limit: 3.0 C. Rebake overshoot limit: 1.5 C.

Peak uniformity is hottest peak minus coolest peak across the batch. Limit: 4.0 C.

Disposition logic:

1. release when ramp, hold, overshoot, uniformity all pass and door_open_alarm = 0.

2. rebake only when the only thermal failure is short hold, rebake_allowed = yes, rebake_count = 0, door_open_alarm = 0.

3. otherwise hold.

Reason code order: warmup_ramp_high, effective_hold_short, effective_hold_long, peak_overshoot_high, uniformity_high.

Rebake profile code: RBK-150C-12M.